

Automated Segmentation of Individual Trees from Terrestrial Laser Scanning (TLS) Data in Australian Native Forest Plots

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INFO6003 Postgraduate Research Project

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1 Background

Terrestrial laser scanning (TLS) captures high-resolution three-dimensional information about forest structure and has become an important tool for estimating tree diameter, height, crown dimensions and woody volume without destructive sampling [Newnham et al., 2015, Calders et al., 2020]. Most tree-level applications, however, first require a plot-level point cloud to be separated into individual trees. This instance-segmentation step remains one of the most difficult parts of automated TLS processing because neighbouring crowns overlap, understorey vegetation obscures stems, and tree architecture varies among forest types [Martin-Ducup et al., 2021, Cherlet et al., 2026].

Existing approaches fall into two broad families. Geometry- and graph-based methods—such as treeseg [Burt et al., 2019], TLS2trees [Wilkes et al., 2023], treeiso [Xi and Hopkinson, 2022] and Rayextract [Lowe and Stepanas, 2021]—are interpretable and can be applied without labelled training data, but their performance depends on assumptions and parameters that may not transfer well between forests. Deep-learning models such as ForestFormer3D, TreeLearn [Henrich et al., 2024] or SegmentAnyTree [Wielgosz et al., 2024] learn patterns directly from data and need little tuning at inference, but they require suitable labelled examples and may perform poorly on unfamiliar forest structures. Related segmentation methods and benchmarks have also been developed across airborne, UAV and personal laser scanning platforms [Wang, 2020, Tockner et al., 2022, Puliti et al., 2023, Xiang et al., 2024], underlining that transferability between sensors and forest types is a central and unresolved challenge.

Cherlet et al. [2026] established a standardised benchmark for TLS tree-instance segmentation across four forest types, using 2,983 manually labelled trees. Their results showed that segmentation performance varied strongly across forest types, and that Rayextract was generally among the best-performing methods, outperforming the deep-learning models on most plots. The study also demonstrated the value of manually corrected reference labels, standard evaluation metrics, and practical measures such as processing time. Notably, the benchmark includes two Australian sites—Litchfield, classified as tropical savanna, and Robson Creek, tropical rainforest—but neither represents the temperate sclerophyll forests of south-eastern Australia.

Segmentation quality directly affects every later tree-level measurement. A merged prediction can combine the stems and crowns of neighbouring trees, while an over-segmented prediction can divide one tree into several false individuals. Either error changes tree counts and can distort estimated crown dimensions, diameter at breast height (DBH), wood volume and biomass. A useful evaluation must therefore consider both whether trees are detected as separate instances and how accurately the points of each detected tree are assigned. It should also report the human effort required to correct the output, because a method that produces reasonable numerical scores may still be impractical if extensive manual editing is needed.

The Blue Mountains monitoring network comprises 36 plots spanning dry sclerophyll, wet sclerophyll and rainforest communities, of which approximately 34 have been scanned with TLS by collaborators at the University of Ghent, alongside species and stem-diameter records. This creates an opportunity to test an established segmentation workflow in forest structures that are not directly represented in the published benchmark. These forests present complex eucalypt architecture, overlapping canopies, multi-stemmed and lignotuberous growth forms and dense understorey, all of which may challenge methods developed elsewhere.

The immediate research need is therefore not to build a new automated model within one semester, but to establish a reliable local workflow, produce a quality-controlled reference dataset, and assess how well an existing automated method performs on these data. Once individual trees have been segmented, TLS can also support estimation of DBH, woody volume, and above-ground biomass (AGB) through quantitative structure models (QSMs). These estimates can be compared with field measurements and inventory-based allometric estimates, while recognising that both approaches carry uncertainty [Raumonen et al., 2013, Calders et al., 2015, Disney et al., 2018].

The proposed contribution is deliberately local and practical: a carefully documented pilot dataset and workflow that can support later comparison across plots, future machine-learning experiments and ecological analysis. The project will focus on one selected plot, or a clearly defined substantial area if the full plot is too large for the available time, and will not claim that results from a single plot generalise to all Australian forests. Greater automation, a second forest type and phone-based photogrammetry remain optional extensions rather than core requirements.

2 Objective

The overall aim is to develop and document a reproducible semi-automated workflow for segmenting individual trees from TLS data in one selected Australian native forest plot, create a quality-controlled reference dataset, evaluate the performance and practical value of Rayextract, and compare TLS-derived tree measurements with available field inventory data. The specific objectives (the intended outcomes, not the tasks) are given in Table 1.

Table 1. Project objectives, priority and the research questions each addresses.

#	Objective	Priority	Addr.
1	To establish the data-quality and preprocessing requirements necessary for reproducible individual-tree segmentation of the selected TLS dataset.	Enabling	–
2	Produce quality-controlled individual-tree reference labels using RayCloudTools and manual correction in CloudCompare, recording throughput and relating per-tree correction effort to structural attributes.	**Core**	RQ1
3	Evaluate how closely Rayextract reproduces the corrected reference labels using instance- and point-level measures, and characterise where segmentation fails by structure and growth form.	**Core**	RQ2, H1
4	Derive DBH and, where data quality permits, QSM-based woody volume and AGB; validate DBH against field measurement and compare AGB with inventory-based estimates.	Secondary	RQ3, H2

3 Hypothesis/question

The project is organised around three research questions and two working expectations, each related to the objectives above.

RQ1.
How efficiently can the RayCloudTools–CloudCompare workflow produce quality-controlled individual-tree labels, and which tree and stand characteristics are associated with greater manual correction effort?

RQ2.
How accurately does Rayextract reproduce the corrected reference labels, how much manual correction does it save, and which forest structures or tree forms are associated with segmentation errors?

RQ3.
How closely do TLS-derived DBH and QSM-based AGB agree with field-measured DBH and inventory-based allometric AGB estimates?

Working expectations

H1. Rayextract performance will decrease in areas of greater local structural complexity, including dense understorey, overlapping crowns and multi-stemmed growth forms. This will be assessed by comparing segmentation scores across structural categories; the expectation is not supported if the difference between categories is negligible relative to its confidence interval.

H2. TLS-derived DBH will show low systematic bias relative to field measurements, reported as mean bias with 95% confidence intervals and Bland–Altman limits of agreement. QSM-based AGB is expected to be positively associated with inventory-based allometric estimates, although systematic differences may remain because the two methods rest on different assumptions.

4 Methodology

4.1 Design

The study will use existing TLS point clouds and associated inventory data supplied by the supervisory team; no new primary data collection is required for the core project. One plot will be selected after considering forest type, point-cloud quality, availability of field measurements, and expected manual workload. If full-plot labelling is not feasible, a fixed and clearly documented area will be used instead.

The analysis operates at three levels: points are used to calculate segmentation overlap; individual trees are used for detection, DBH and biomass comparisons; and local areas or subplots are used to describe structural complexity. Plot selection is treated as an explicit methodological decision, and the selected boundary, minimum tree criteria and any exclusions will be documented before evaluation begins. Field records are georeferenced only to the quadrant level (NE/SE/SW/NW); no within-plot tree coordinates exist. Inventory trees will therefore be matched to segmented trees by constrained assignment within each quadrant, using species, stem count and stem-diameter signature, with unresolved cases flagged rather than forced. The final sample size will be confirmed by the initial audit, since it depends on the stem density of the selected plot. The overall processing and evaluation pipeline is summarised in Figure 1.

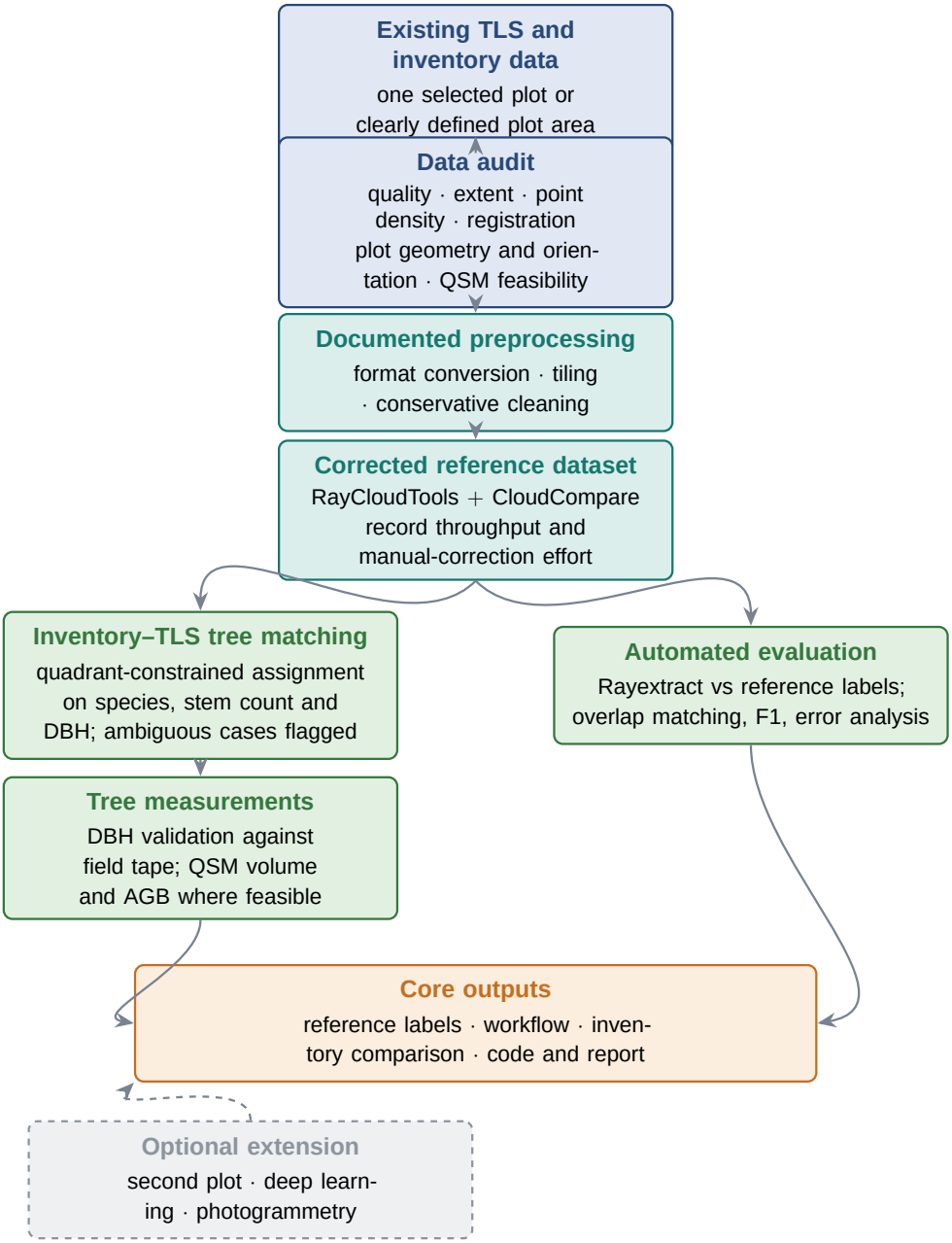


Figure 1. Processing and evaluation pipeline for the proposed workflow.

4.2 Data Needs and Fixed Parameters

- **Unit of analysis:** the individual tree, as both a segmentation output and a reference instance.
- **Input:** existing TLS point clouds from Blue Mountains monitoring plots, scanned by University of Ghent collaborators and held by the supervisory group. No new primary data collection. One plot is selected for the core work; forest type is recorded so that structural difficulty can be interpreted.
- **Reference:** individually segmented trees produced in the reference-dataset construction step. Co-located field inventory (species identity, stem diameter, demographic records) provides the ground truth for biomass and allometric validation.
- **Covariates:** *per plot*—stem density, basal area, terrain slope, number of scan positions, registration residual, understorey point density (returns between 0.5 and 3 m). *Per tree*—height, DBH, crown extent, multi-stem flag, distance to nearest scan position.
- **Fixed parameters:** subplot split 60/20/20; edge trees ($>$90% overlap with the test subplot) retained but excluded from metrics; IoU 0.5 for a correct instance; Rayextract at published defaults, with `height_min` and `max_diameter` set from the training data; DBH at 1.3 m by RANSAC circle fit on a 0.10 m slice (residual < 0.02 m); QSM with ten averaged fits per tree; AGB $=$ woody volume $\times$ species-specific wood density; random seed 42 with deterministic PyTorch settings [following Cherlet et al., 2026]. Matching is constrained to quadrant, DBH tolerance ± 2 cm, stem-count agreement required; log, stump and standing-dead records are excluded from the matched comparison.

4.3 Data Audit and Preprocessing

The first stage will document file formats, plot extent, point count, point density, registration quality, scan positions, and the completeness of matching field inventory. The audit will also determine whether leaf-on conditions or occlusions are likely to limit QSM fitting. Preprocessing will include format conversion, tiling, conservative noise filtering, ground filtering (for example, cloth-simulation filtering; Zhang et al., 2016) and conversion to raycloud format where required, implemented as a scripted workflow (PDAL, laspy) so that it can be re-run reproducibly. The same processed input will be used for both reference labelling and automated evaluation, and points not assigned to a reference tree will remain as background so that the evaluation environment is realistic.

The audit will also compare the dates and definitions used in the TLS and field datasets. Differences in measurement year, DBH protocol, treatment of multi-stemmed individuals or recorded mortality may produce disagreement unrelated to the segmentation algorithm. These issues will be recorded in a data dictionary and considered when interpreting results. A small representative sample will be processed first so that file sizes, processing time and likely labelling effort can be estimated before the core area is fixed.

4.4 Reference Dataset Construction

RayCloudTools will generate initial tree candidates, followed by manual inspection and correction in CloudCompare. Trees that are merged, split or contaminated by neighbouring vegetation will be corrected until the selected area has a consistent level of detail. Following the protocol of Cherlet et al. [2026], trees are clipped into separate clouds, and no points are deleted at any stage, so that the evaluation environment remains realistic. The workflow

will be developed with training and guidance from Erick Calderon-Morales, and a sample of completed labels will be reviewed with an experienced team member to check consistency.

A written labelling protocol will define how tree points are distinguished from ground, understorey and neighbouring crowns, and how multi-stemmed or ambiguous individuals are represented. Uncertain cases will be marked and discussed with the supervisory team so that final labels reflect a consistent ecological interpretation rather than ad hoc decisions. Quality control will include a second inspection pass and targeted review of difficult trees. The project will record trees completed per operator-hour, the proportion requiring substantial correction, and—for each tree—an ordinal correction-effort class (none / minor / substantial) together with the correction type. Effort will then be related to per-tree attributes recorded in Section 4.2 (DBH, height, crown extent, multi-stem or lignotuberous form, local stem density, understorey point density, distance to nearest scan position) so that the analysis identifies *which* trees drive labelling cost rather than only cataloguing correction types.

4.5 Automated Segmentation Methods

Rayextract will be the primary automated method because it is part of the existing workflow and performed strongly in the benchmark of Cherlet et al. [2026]. Settings will follow published guidance [Lowe and Stepanas, 2021], with plot-specific values such as minimum tree height and maximum diameter set from the data audit. Any parameter selection will use a development area kept separate from the final evaluation area, so that reported performance is not tuned on the same trees.

Evaluation will follow the framework of Cherlet et al. [2026]. Predicted trees will be matched to reference trees by point-cloud overlap using an optimal assignment procedure, with an Intersection over Union (IoU) threshold of 0.5 defining a successful detection, consistent with the published benchmark. Instance-level precision, recall, and F1 will describe whether the correct number of trees is detected, while point-level precision, recall and F1 for matched trees will describe how well the extent of each tree is delineated. Both are reported because they answer different questions: a method can find the right number of trees while assigning their points poorly, or vice versa. Boundary-crossing trees will remain in the point cloud but be excluded from metric calculation where too much of the tree lies outside the evaluation area. Runtime and manual-correction time will also be recorded, so that accuracy is interpreted alongside practical cost rather than a single ranking score. The methods and their configuration are summarised in Table 2.

Runtime, peak memory, and human-intervention time are recorded per plot per method, so practical cost is reported alongside accuracy.

4.6 Tree Measurement and Biomass Comparison

DBH will be estimated from each segmented tree at 1.3 m, matching the field-inventory protocol, by robust (RANSAC) circle fitting to a short horizontal stem slice, with poor fits rejected on a residual threshold set during a pilot analysis. Cylinder fitting will be considered for strongly non-circular stems, which are common in this forest type.

Because field records carry quadrant membership but not within-plot tree coordinates, TLS trees will first be assigned to quadrants using the plot centre and orientation, then matched to inventory records within each quadrant by optimal assignment (Hungarian algorithm; Kuhn, 2005), with a cost combining DBH difference, stem-count agreement and species plausibility. The diameter tolerance accommodates both fitting error and growth

Table 2. Segmentation methods, their role in the study, and configuration.

Role	Method	Settings
Geometric *(primary)*	Rayextract — RayCloudTools suite [Lowe and Stepanas, 2021]	Published defaults; `height_min` and `max_diameter` set from the training/validation data as in §4.2
Deep learning *(optional)*	ForestFormer3D pretrained weights (SegmentAnyTree/TreeLearn as fallback), inference only	As published; inference only unless the fine-tuning stretch objective is reached
Baseline	In-house stem detection plus vertical region growing	Fixed, untuned
Stretch	FSCT [Krisanski et al., 2021], TLS2trees [Wilkes et al., 2023]	Only if the core benchmark completes by Week 10

between the field campaign and the scan. Roughly a quarter of trees share a quadrant with a same-species individual of similar diameter (Figure 2), so ambiguous assignments will be flagged and excluded from per-tree statistics, with the achieved match rate reported and distribution-level comparison (quadrant DBH distribution, basal area, stem count) used where matching is incomplete. Logs, stumps and records without a diameter are excluded; standing-dead stems are retained for DBH validation but excluded from AGB comparison.

Roughly a quarter of inventory records are multi-stemmed, with up to eleven stems. TLS and field values will be compared stem-by-stem where stems can be resolved and aggregated to a quadratic mean diameter $D = \sqrt{\sum d_i^2}$, which preserves total basal area, for whole-tree comparison and allometric AGB.

Where point-cloud quality is sufficient, TreeQSM will estimate woody volume [Raumonen et al., 2013], with repeated fits per tree averaged to assess reconstruction variability. AGB will be calculated from woody volume and species- or genus-level wood density, then compared with allometric estimates from the field inventory; allometric values are a reference point, not biomass ground truth. Occlusion, foliage, missing upper branches and segmentation errors all affect reconstruction, so diagnostics and visual checks will identify unreliable trees. Results may be reported separately for single-stemmed and structurally complex trees, and the biomass component is restricted to a high-quality subset if whole-plot reconstruction is not defensible.

4.7 Statistical Analysis

- **Agreement (H2, RQ3).** Bias, RMSE and Bland–Altman limits of agreement for DBH where sufficient matched trees are available; paired bootstrap resampling with 95% confidence intervals for differences between QSM-based and allometry-based AGB, reported at tree and plot or sampled-area level.
- **Segmentation performance (H1, RQ2).** Instance- and point-level scores summarised overall and, where sample size permits, compared across structural categories such as local stem density, tree height and multi-stemmed growth form.
- **Interpretation.** Emphasis on effect sizes, uncertainty and practical interpretation rather

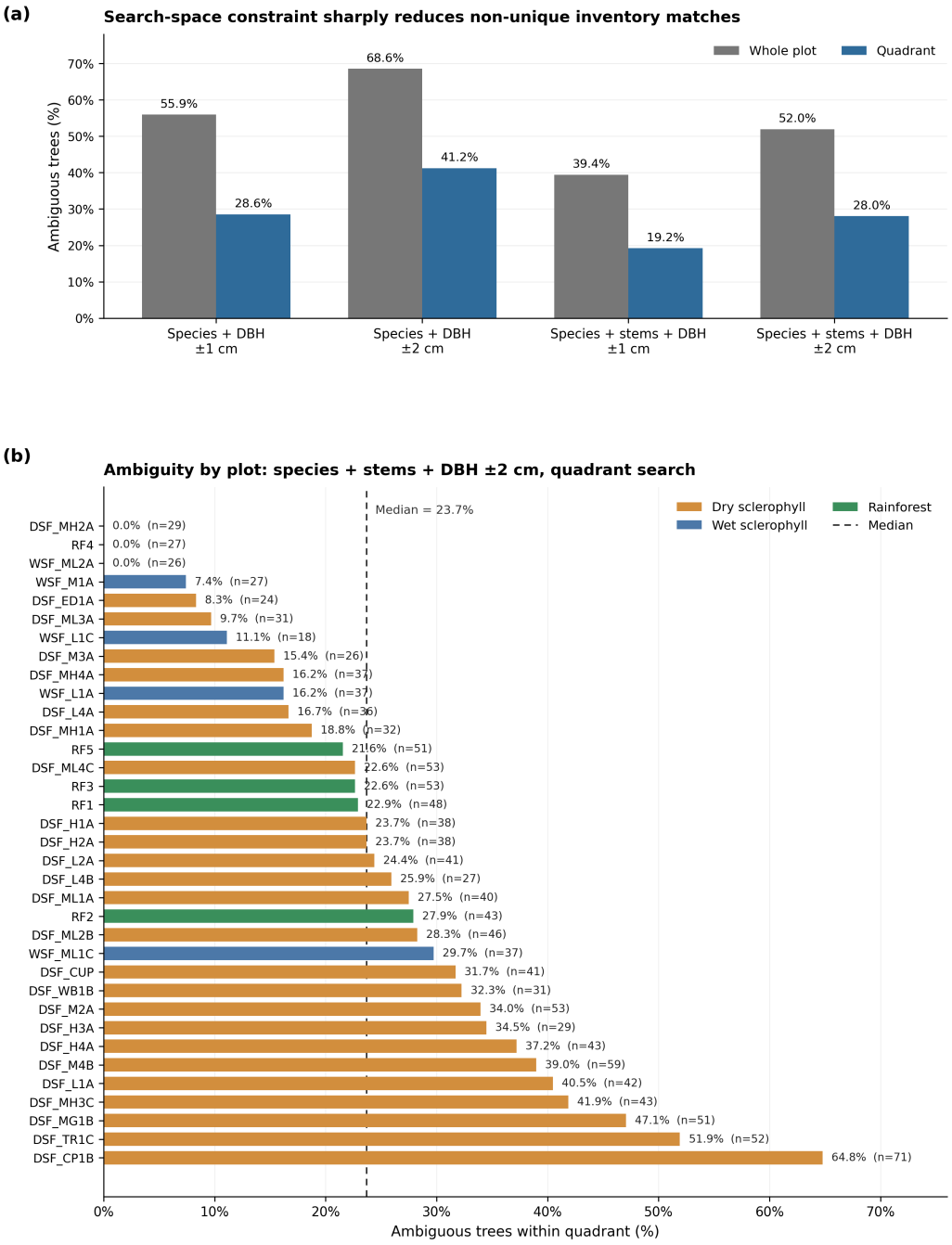


Figure 2. Ambiguity of inventory-to-TLS tree matching. **(a)** Proportion of inventory trees sharing their search space with an indistinguishable record, under four matching criteria, with and without the quadrant constraint. **(b)** Per-plot ambiguity under species + stem count + DBH ± 2 cm within quadrant, ranked and coloured by forest type; n = standing inventory trees.

than significance testing alone, with accuracy read alongside runtime and manual-correction effort.

- **Choice of model.** The final statistical approach is confirmed once the number and distribution of observations are known; more complex models, such as mixed-effects models accounting for subplot grouping, are used only if the data support them.
- **Correction effort (RQ1).** Correction-effort class modelled against per-tree structural covariates (ordinal or binary regression, or descriptive cross-tabulation if numbers are small), reporting effect sizes with confidence intervals.

4.8 Error Analysis

Representative segmentation failures will be classified as omission, false detection, merge, split, crown truncation, base misattribution or understorey contamination, and cross-tabulated by growth form where numbers allow. Each case will be documented with rendered views of the input, the prediction, and the reference tree, so that the analysis explains why errors occurred rather than relying on numerical scores alone.

4.9 Reproducibility

Code and configuration files will be maintained in a GitHub repository with regular commits, in line with subject requirements. Analysis and reporting will be developed as literate-programming documents (Quarto/Jupyter) so that figures, tables and metrics regenerate directly from code. The computational environment will be captured with a declarative package manager (pixi) producing a lock file, so the workflow can be reproduced exactly on the project workstation and by the supervisory group. Setup will be supported by Erick Calderon-Morales. Processing steps, software versions, random seeds, data exclusions and parameter decisions will be documented; raw data will remain read-only; and derived outputs will use consistent identifiers for linking predictions, reference trees and inventory records.

4.10 Ethical Consideration and Safety

No human or animal participants are involved, and no ethics approval is anticipated. The blinded visual assessment involves colleagues rating anonymised outputs and collects no personal data. Point-cloud data are large; raw data will be retained read-only with structured backup, and code held under version control. Should any fieldwork occur, standard risk assessment and site-access permissions would apply.

5 Expected outcomes

The core project is expected to deliver:

1. a quality-controlled set of individual-tree reference labels for one selected plot or defined plot area;
2. a documented and reproducible RayCloudTools–CloudCompare workflow, including throughput, correction effort, and the structural conditions under which manual correction is required;

3. an evaluation of Rayextract using standard accuracy metrics, practical processing cost and structured error analysis;
4. TLS-derived DBH and, where feasible, QSM-based woody volume and AGB compared with field-inventory estimates; and
5. organised code, configuration files, and documentation suitable for continued work by the research group.

The main contribution is a credible local workflow and reference dataset for Australian native forest structure, in a forest type absent from the published benchmark. Even if the optional machine-learning work is not reached, the core outputs will support future algorithm development, additional plot labelling, and more reliable comparison of TLS with conventional forest inventory.

6 Program of work

The project is designed for the one-semester INFO6003 timeline. Existing scans, field inventory, open-source software, a high-performance workstation and direct workflow training reduce start-up risk. The manual labelling workload is controlled by limiting the core scope to one plot or a fixed area, and automated processing and QSM fitting can run in batches while report writing proceeds. The timeline and task dependencies are shown in Figure 3.

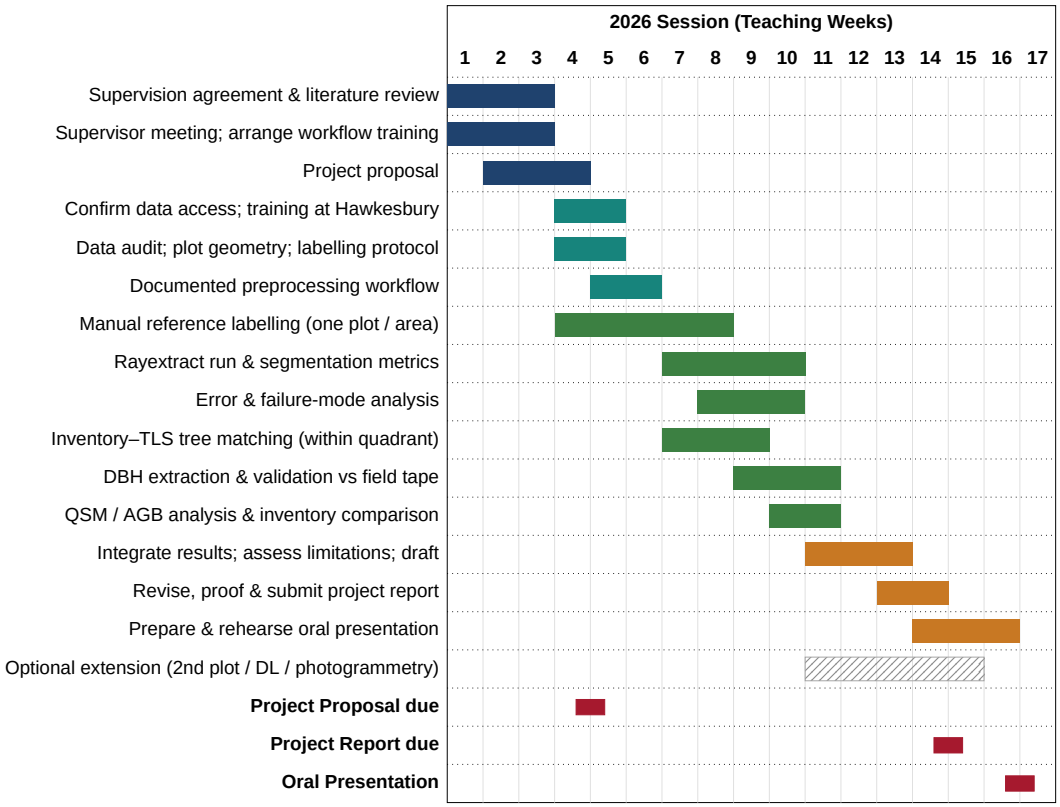


Figure 3. Gantt chart of the project timeline and tasks across the 2026 teaching session. Colour groups: **navy** set-up & proposal; **teal** data preparation; **green** labelling, evaluation & measurement; **orange** writing & dissemination; hatched optional extension; ♦ assessment milestones.

6.1 Feasibility

The compressed timeline is the dominant constraint, and the plan is built to absorb it. The data already exists, so there is no collection phase. The core tools are open source, and the operational workflow is already in use by the supervisory group, with hands-on training provided by Erick Calderon-Morales. The expensive step—manual labelling—is capped at one plot and front-loaded. Biomass analysis and automated processing are batch work that can run unattended on the provided high-performance workstation.

6.2 Risk Management

The principal risks to delivery, and their planned mitigations, are set out in Table 3.

Table 3. Key project risks, their likelihood and impact, and planned mitigation.

Risk	Likelihood	Impact	Mitigation
Data access delayed, or poor registration / large occlusion gaps in the selected plot	Medium	High	Confirm access and inspect registration quality during the Week 5 audit (§4.3); select an alternative plot or a smaller defined area; if access is delayed, run the same workflow on the Cherlet et al. [2026] benchmark data while it is resolved.
Inventory trees cannot be reliably matched to TLS trees	Medium	High	Field data are quadrant-level only; matching relies on species, stem count and diameter within quadrant. Confirm plot geometry and orientation during the Week 5 audit; report match rate; fall back to quadrant-level distributional comparison where per-tree matching is ambiguous.
Wood density or allometric equations unavailable for some species	Medium	Medium	Use genus- or community-level density and published Australian allometry; report biomass with explicit uncertainty; flag species with weak coverage.
Leaf-on scans or occlusion limit QSM reconstruction	Medium	Medium	Assess QSM feasibility in the data audit; test leaf–wood separation on a subset; restrict biomass analysis to trees with acceptable reconstruction diagnostics and report single- and multi-stemmed trees separately.

Table 3 (continued)

Risk	Likelihood	Impact	Mitigation
Compute, storage or software problems on large point clouds	Medium	Medium	Tile and subsample large clouds; stage processing in batches; keep the in-house baseline as a floor and maintain structured backups.
Workstation / GPU access or booking delays	Medium	Medium	Request and book HPC early through supervisors; tile and subsample large clouds; keep the in-house baseline as a floor.
Manual reference labelling takes longer than budgeted	High	Medium	Cap the core scope at one plot or a fixed defined area; label a fixed area rather than a fixed tree count; review throughput weekly and prioritise complete quality control over expansion.
Machine-learning or photogrammetry stretch work not reached	Medium	Low	Scoped as optional and scheduled last; the reference dataset, evaluation and inventory comparison stand alone as a complete outcome.
Supervisor availability / scope change	Medium	Medium	Confirm scope at the first meeting; maintain regular check-ins and a written record of agreed decisions.

6.3 Resources

- **Data:** existing TLS point clouds (University of Ghent) and matched field inventory held by the supervisory group; institutional storage for large point-cloud datasets, raw data read-only and backed up. Public benchmark datasets as the contingency.
- **Computing:** a dedicated high-performance workstation with sufficient GPU capability, provided by the project team with remote access for the student; booking coordinated through Prof. Choat.
- **Software:** all core tools open-source—Python (laspy, PDAL, Open3D, scikit-learn, PyTorch), R (lidR, lme4), RayCloudTools (Rayextract), CloudCompare, TreeQSM for quantitative structure modelling, and Git.
- **People:** Erick Calderon-Morales for workflow training and ongoing support; a small number of colleagues for any blinded visual assessment.
- **Stretch only:** smartphone RGB camera, structured image-acquisition protocol, and structure-from-motion or multi-view stereo software such as COLMAP.
- **Training:** anticipated development in point-cloud libraries, QSM and biomass workflows, GPU workflows and mixed-effects modelling, supported by the in-person Hawkes-

bury session, institutional training and expertise within the supervisory group.

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